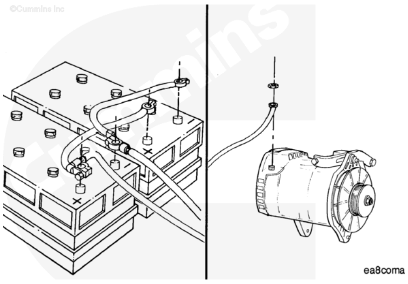
Trainings

Remove

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



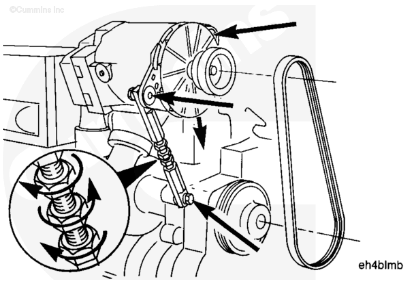
- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the wiring and ground strap from the alternator. See equipment manufacturer service information.

Loosen the adjusting link and the alternator mounting capscrews.

Note : The lower jam nut has left-hand threads.

Loosen both of the jam nuts. Turn the adjusting screw to relieve the belt tension.

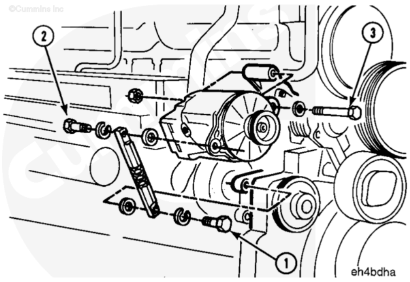
Remove the alternator belt.



Remove capscrews (1) and (2) and the adjusting link.

Remove capscrew (3) and nut.

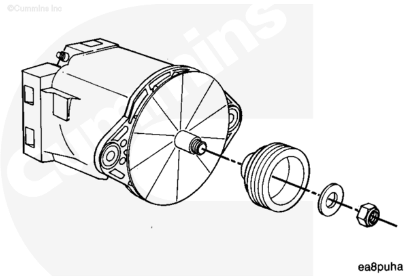
Remove the alternator.



Inspect for Reuse

Remove the nut and the pulley from the alternator.

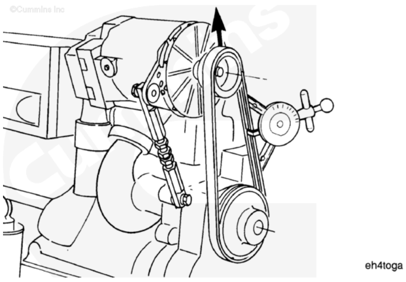
Clean and check the pulley for reuse.



Test

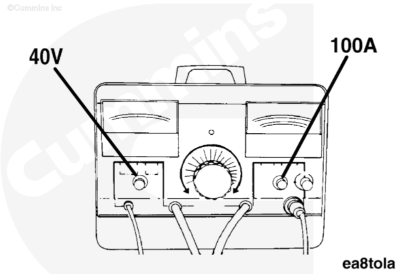
The following instructions are for use with the Part Number 3377193 Inductive Charging and Cranking System Analyzer, or equivalent.

Note : Before performing the following test, be sure the alternator belt is tightened to the correct specifications. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/20/20-013-005-tr.html).



Set the voltage selector knob to the appropriate scale. For a 24 volt system, choose the 40 volt scale.

Set the amp selector knob to 100 amps.



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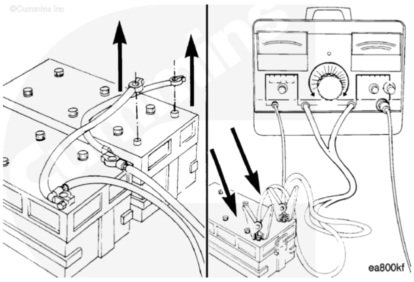
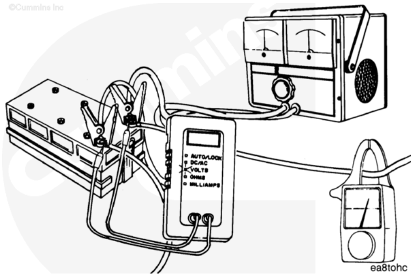


Remove the cables to any other battery in the circuit.

Connect the correct analyzer leads to the **positive** and **negative** terminals on the battery.

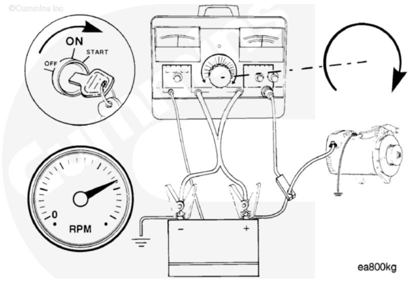
Connect the clamp-on amp pick-up to the alternator output cable as far away from the alternator as possible.



Operate the engine at high idle and turn the analyzer load control knob **clockwise** until a maximum amps reading is obtained.

Note : Do not let the load volts drop below 26 volts for a 24 volt system.

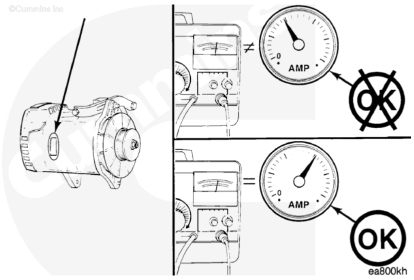
The maximum amp reading is the alternator output, and **must** meet the alternator manufacturer's specifications.



Note : The alternator maximum rated output is normally stamped or labeled on the alternator.

Note : Also check the equipment ammeter gauge. If it does not read approximately the same as the test equipment, it should be replaced.

Note : If the alternator output is not within 10 percent of rated output, repair or replace the alternator. See equipment manufacturer service information.



Turn the analyzer load control knob **counterclockwise** to the “OFF” position and shut off the engine.

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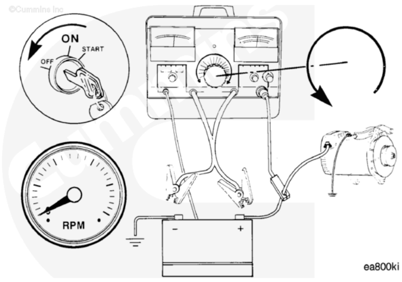


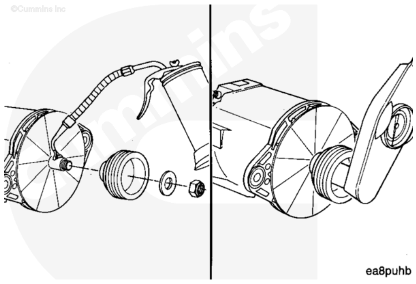
Remove the test equipment. Connect all battery cables that were removed.

Install

Lubricate the shaft with engine oil. Install the pulley and nut on the alternator shaft.

Tighten the nut.

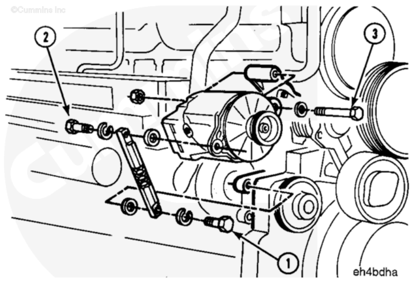
Torque Value: 100 n•m [75 ft-lb]



The belt **must** be adjusted before the capscrews are tightened.

Note : The end of the adjusting link with the largest area at the capscrew hole must be nearest to the alternator.

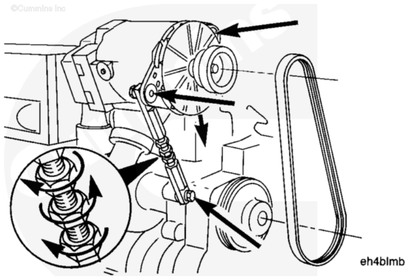
Install the alternator and the adjusting link as shown.



⚠ CAUTION ⚠

Do not attempt to pry the belt on the pulley to avoid damage to pulley and belt.

Install the belt. Turn the adjusting screw **counterclockwise** to shorten the link, if necessary.

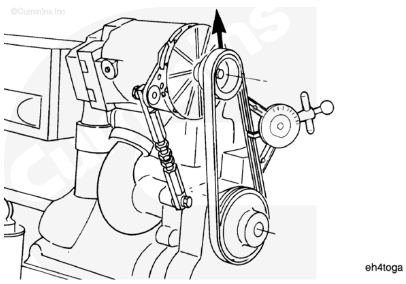


Turn the adjusting screw **clockwise** to tighten the belt.

Tighten the belt.

Measurements		
	n	lbf
Belt Tension	670	150

Use the (Burroughs) belt tension gauge, Part Number ST-1138, to check the belt tension.

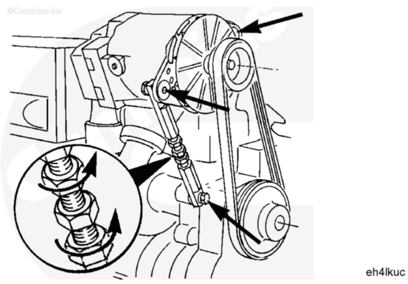


Note : The lower jam nut has left-hand threads.

Tighten the jam nuts on the adjusting screw.

Tighten the adjusting link and alternator mounting capscrews.

Torque Value:



Jam Nuts

1. 55 n•m [40 ft-lb]

Torque Value:

Alternator Mounting Capscrews

1. 55 n•m [40 ft-lb]

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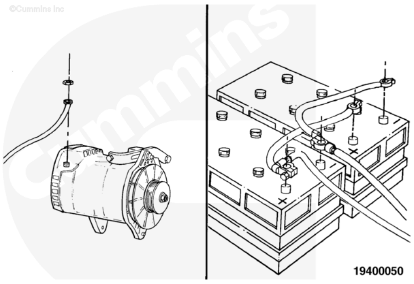

- Connect the wiring to the alternator. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.